

ELASTICITY

Stress and Strain:

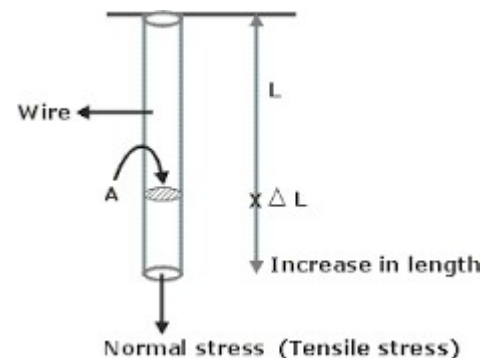
When a body is subjected to deforming force, a restoring force is developed in the body. The restoring force per unit area is known as **stress**. This restoring force is equal in magnitude but opposite in direction to the applied force.

- If F is the force applied and A is the area of cross section of the body, then $\text{Stress} = F/A$.
- The SI unit of stress is Nm^{-2} or pascal (Pa) and its dimensional formula is $[ML^{-1}T^{-2}]$.

The ratio of change in dimension of a body to the original dimension is called **strain**. Strain has no units or dimensions.

There are three types of stress and strain.

- 1) **Longitudinal stress:** If a force is applied normal to the cross section of the body, then the restoring force developed per unit area is called longitudinal stress. Longitudinal stress may be of two types i.e. tensile stress or compressive stress i.e. it can cause either extension (tensile stress) or compression (compressive stress) depending on the direction. In both cases, there is change in length of the body.

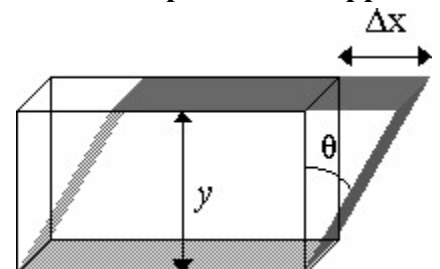


Longitudinal strain: The ratio of change in the length (ΔL) to the original length (L) of the body is known as longitudinal strain.

i.e. **Longitudinal strain** = $\frac{\Delta L}{L}$

- 2) **Shearing Stress:** If a deforming force is applied parallel to the body's surface area, there is relative displacement between opposite faces. The restoring force per unit area developed due to applied tangential force is known as tangential or shearing stress.

Shearing strain: As a result of applied tangential force, there is relative displacement Δx between opposite faces of the body as



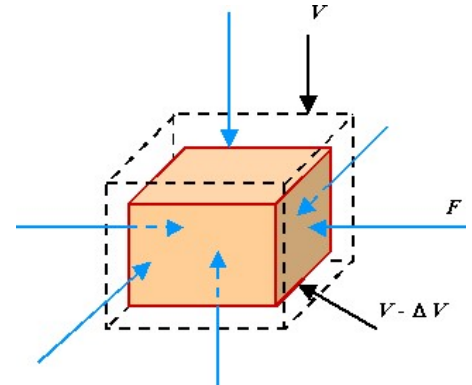
shown in figure. The strain so produced is known as shearing strain.

Shearing strain is defined as the ratio of relative displacement of the faces Δx to the height 'h'.

$$\text{i.e. Shearing strain} = \tan\theta = \frac{\Delta x}{y}$$

where θ is the angular displacement of the surface from the vertical (original position). Usually θ is very small and hence $\tan\theta \approx \theta$.

- 3) **Normal stress or Bulk stress:** If a force acts in perpendicular direction at each point of the surface then restoring force per unit area is called bulk stress or normal stress. In this case, only volume of the body changes without any change of its shape. **Volumetric strain or Bulk strain:** It is defined as the change in volume (ΔV) to the original volume (V), without any change in shape.



$$\text{i.e. Volumetric strain} = \frac{\Delta V}{V}$$

Hooke's Law: It states that Stress is directly proportional to strain within elastic limit.

$$\text{i.e. stress } \propto \text{strain or } \frac{\text{stress}}{\text{strain}} = \text{constant}$$

The ratio of stress to strain is called the modulus of elasticity or elastic constant. It is a characteristic property of the material. There are three types of moduli.

1. Young's Modulus (Y):

Young's Modulus is defined as the ratio of longitudinal (linear) stress to longitudinal strain within the elastic limit.

Since strain is a dimensionless quantity, the unit of Young's Modulus is the same as that of stress

$$\text{i.e. } Y = \frac{F/A}{\Delta l/L} \quad \text{Nm}^{-2}.$$

F represents the force applied normal to the area 'A' of a wire of length 'L' and ' Δl ' is the change in

length. For metals like iron, steel, copper, aluminium etc., the Young's moduli are very large. Therefore, these materials require a large force to produce small change in length.

2. Rigidity modulus(n):

The ratio of shearing stress to the corresponding shearing strain is called the shear modulus or

Rigidity modulus of the material. $n = \frac{F}{A\theta} \text{ Nm}^{-2}$

F is the tangential force and θ is the shear strain.

3. Bulk Modulus (K):

Bulk Modulus is defined as the ratio of normal stress to volumetric strain with in the elastic limit.

$$K = \frac{FV}{A(\Delta v)} \text{ Nm}^{-2}$$

F/A is the normal stress and Δv is the change in volume and V is the original volume.

Poisson's ratio, σ , is the ratio of lateral strain to longitudinal strain. $\sigma = \beta/\alpha$

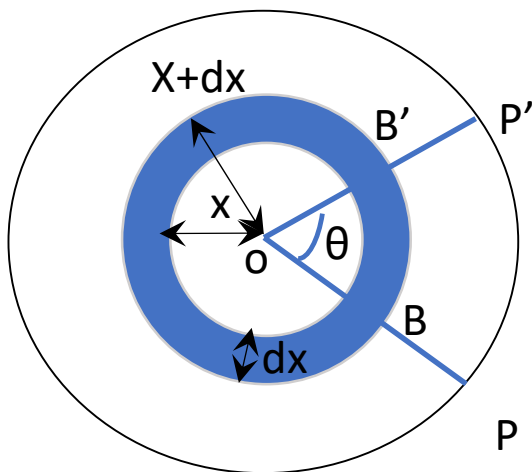
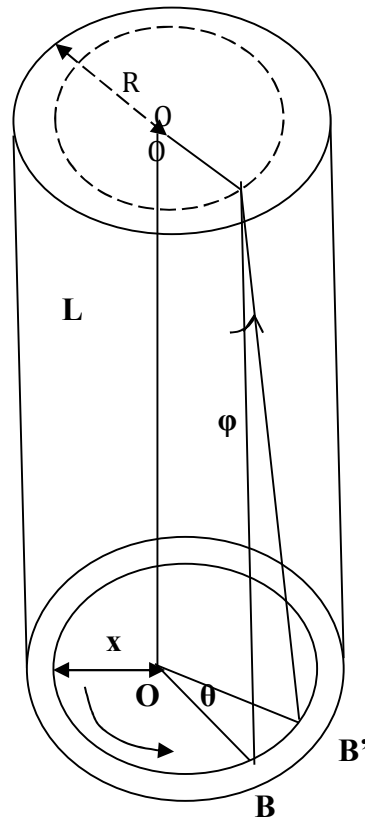
α is increase in length per unit length per unit stress in the direction of the stress and

β is decrease in length per unit length per unit stress in a direction perpendicular to the stress.

i.e, β is lateral strain per unit stress and α is the longitudinal strain per unit stress.

➤ Torsion of a Cylinder:

Consider a cylinder of length 'L' and radius 'R'. Let the cylinder be clamped at the upper end a twisting couple be applied at the lower end. Let θ be the angle of twist.



$$\begin{aligned} \text{Area of the shell} &= \pi(x + dx)^2 - \pi x^2 \\ &= 2\pi x dx \quad ; \text{ignoring } dx^2 \end{aligned}$$

The cylinder can be considered as made up of a number of co-axial hollow cylinders of varying radii (0 to R).

Term: 01-06-2023 to 09-09-2023

At the bottom end, each of these radii are twisted through an angle, θ due to the applied external couple. As a result, OB is displaced to OB' and OP is displaced to OP' without change in dimensions.

A line AB parallel to OO' is displaced to AB' through an angle ϕ called the angle of shear. This is an example of pure shear as there is no change in either length or radius of the cylinder.

We can calculate the twisting couple on the co-axial cylinder of radius OB (OB = x) and integrate the expression between the limits, x=0 and x=R to obtain the magnitude of the twisting couple on the cylinder. We can write from the geometry of the figure,

$$BB' = x \theta = L\phi$$
$$\text{Or, } \phi = x \theta / L \quad \text{-----(i)}$$

If δF is the tangential force acting on a cylindrical shell of radius, 'x' and thickness, 'dx', then

The tangential stress = $\delta F / \text{area of the shell} = \delta F / 2\pi x dx$ -----(ii)

The rigidity modulus = tangential stress / shear strain

$$n = (\delta F / 2\pi x dx) / \phi \quad \text{-----(iii)}$$

$$\delta F = 2\pi n x dx. \phi \quad \text{----- (iv)}$$

Twisting couple on the hollow cylinder of radius 'x' and thickness, 'dx' can be written as

$$\delta C = \text{tangential force} \times \text{distance} = n. \phi. 2\pi x dx. x \quad \text{----- (v)}$$

n, is the rigidity modulus of the material and is given by the ratio of tangential stress to shear strain.

Substitute for ϕ in eqn. (ii) to obtain

$$\delta C = 2\pi n \theta x^2 dx . x / L = 2\pi n \theta x^3 dx / L \quad \text{----- (vi)}$$

Twisting couple on the solid cylinder of radius 'R',

$$C = \frac{2\pi n \theta}{L} \int_0^R x^3 dx = \frac{2\pi n \theta}{L} [R^4 / 4]$$

$$C = \frac{\pi n R^4 \theta}{2L}$$

When $\theta = 1$ radian, $C = \frac{\pi n R^4}{2L}$ is the couple per unit twist

Torsion Pendulum:

A rigid wire of length 'L' and radius 'r' is fixed at one end and forms the axis of rotation for a regular or an irregular body attached to the free end. When the wire is twisted at the free end, the body is set into oscillations and these oscillations are called torsional oscillations.

Let I be the moment of inertia of the body about the given axis and T be the time period for torsional oscillations. Let the restoring couple per unit twist be 'C' and 'θ' be the angle of twist.

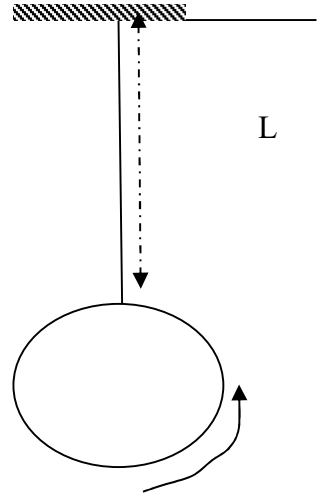
$$I \frac{d^2\theta}{dt^2} = -C \theta$$

$\frac{d^2\theta}{dt^2} + \frac{C}{I} \theta = 0$ represents the simple harmonic equation for torsional oscillations. From the above equation, we can get the time period for torsional oscillations

$$T = 2\pi\sqrt{I/C}$$

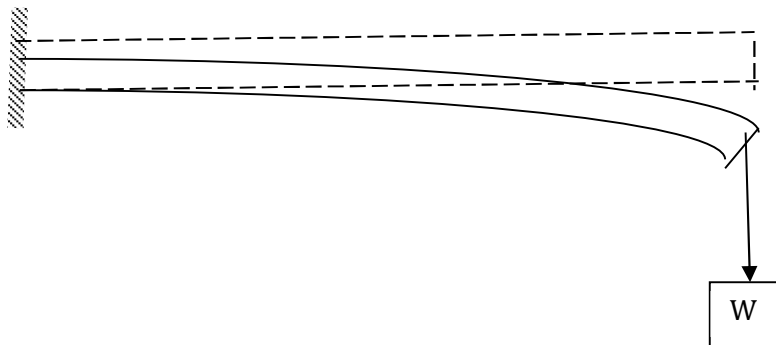
$$\text{Or, } \frac{I}{T^2} = \frac{C}{4\pi^2} \quad \text{where } C = \frac{\pi n r^4 \theta}{2L}$$

For a given wire, C is constant and hence the ratio I/T^2 is constant irrespective of the body or the axis of rotation. This principle is used to determine the moment of inertia of irregular objects.



➤ Bending Moment of a Beam:

A beam is a structural member whose length is very large compared to other dimensions. In the simple theory of bending of beams, the shear stresses are neglected and only tensile or compressive stresses are considered.



Consider a beam fixed at one end and loaded at the other end. Due to the applied force, a force of reaction is

set up at the fixed end of the beam.

These two equal and opposite forces constitute a couple which tends to rotate the beam.

Since the beam is elastic a restoring couple is set up to bring the beam back to equilibrium.

The beam can be imagined as made up of a number of longitudinal filaments. Under the applied load the upper filaments will undergo extensions and assume a convex form and the lower filaments will undergo compressions and assume a concave form. The filament that does not undergo any change in dimension is the **neutral axis**. The magnitude of extension or compression will depend on the distance of the filament from the neutral axis. “**The moment of the restoring couple is called bending moment of a beam**”.

Bending Moment is= YI_g/R ,

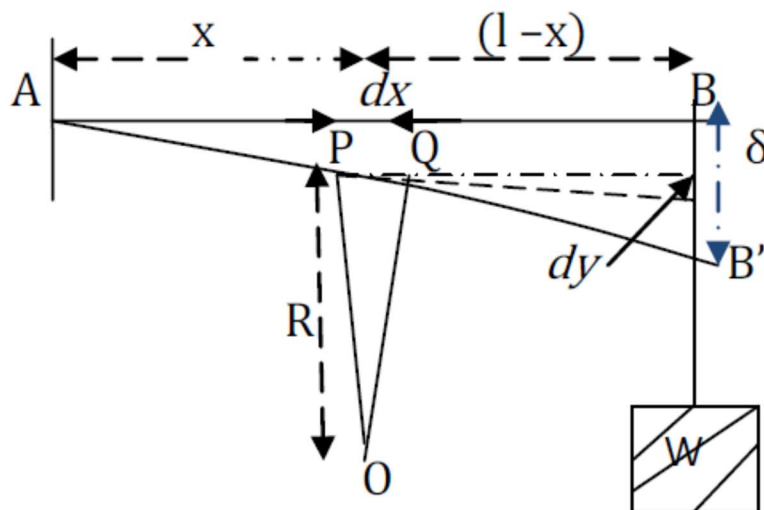
where $I_g = \sum \delta A z^2 = A k^2$ is the geometrical moment of inertia and depends on the area of cross section, ‘A’ of the beam and the radius of gyration k of the area about the neutral axis.

- For rectangular cross section , $I_g = b d^3/12$, b is the breadth and d is the thickness.
- For circular cross section, $I_g = \pi r^4/4$, r is the radius

R is radius of curvature and Y is Young’s modulus.

➤ Expression for Youngs Modulus of a cantilever:

A single cantilever is a beam fixed at one end and loaded at the other end. Consider a load, ‘W’ applied to the free end of the cantilever of length, ‘ l ’. Let δ be the deflection of the free end of the cantilever under the load. The deflection is maximum at the free end and is equal to zero at the fixed end.



Term: 01-06-2023 to 09-09-2023

Let $PQ = dx$, be a small section of the neutral axis AB. PQ is at a distance x from the fixed end. Radius of curvature of neutral axis is $= R$

Deflection of $PQ = dy$.

The radius of curvature of the neutral axis is given by the standard expression

$$\frac{1}{R} = \left[\frac{d^2 y / dx^2}{\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2}} \right] = \frac{d^2 y}{dx^2} \text{ -----(i)}$$

Here, (dy/dx) is very small and therefore higher powers of it can be neglected.

$$\therefore \frac{1}{R} = \frac{d^2 y}{dx^2}$$

The moment of applied force acting on the element $PQ = W(l-x)$ -----(ii)

The restoring couple due to the elasticity of the beam $= YI_g / R$ -----(iii)

At equilibrium, these two are equal and opposite.

$$\therefore YI_g / R = W(l-x)$$

$$\text{i.e. } YI_g \left(\frac{d^2 y}{dx^2} \right) = W(l-x)$$

On integration, we get

$$YI_g \int \frac{d^2 y}{dx^2} dx = w \int (l-x) dx$$

$$YI_g \frac{dy}{dx} = w(lx - x^2/2) + C \text{ where } C \text{ is the constant of integration.}$$

The deflection is zero at $x=0$ and $dy/dx=0$. Substituting this condition, we get $C=0$.

Again on integration, we get

$$YI_g \int_0^\delta dy = w \int_0^l \left\{ lx - \frac{x^2}{2} \right\} dx$$

$$YI_g \delta = w \left[\frac{lx^2}{2} - \frac{x^3}{6} \right]_0^l = w \left[\frac{l^3}{2} - \frac{l^3}{6} \right] = \frac{wl^3}{3}$$

$$\delta = \frac{wl^3}{3YI_g} \text{ -----(v)}$$

For a beam of rectangular cross section, $I_g = b d^3 / 12$. Substituting this we obtain,

$$\delta = \frac{4wl^3}{Yb^3}$$

$\therefore$ Young's Modulus of the cantilever is given by,

$$Y = \frac{4Mgl^3}{bd^3\delta} \quad \text{where } w=Mg$$